

Methods to Combat Climate Change in Oceans on an International Level

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Introduction

Over the past few decades, the impacts of climate change on the world have become increasingly destructive. As described in a 2022 report by the World Meteorological Organization, “[extreme] heatwaves, drought and devastating flooding have affected millions and cost billions [in just one year]” (WMO).¹ Climate change is primarily caused by the burning of fossil fuels and other human activities that increase concentrations of greenhouse gases (NASA).² Its repercussions have been estimated to become irreversible by the year 2030 if society doesn’t put effort towards solving it (United Nations^a).³

Eric Bender proposed in his paper ‘Urban evolution: How species adapt to survive in cities’ that “studies [on urban organisms can] provide important clues [on how creatures respond to climate change]”, thus helping scientists counter the current environmental crisis more effectively (Bender).⁴ He argues that observing organisms can help humanity create and sustain stronger environments not only in the present, but also the future (Bender). This notion of utilizing nature to assist humanity in becoming resilient also appears in Julia Watson’s TEDtalk ‘How to build a resilient future using ancient wisdom’ about using indigenous innovations to reduce the flaws with modern technology, the usage of which is a contributing factor in global

¹ The World Meteorological Organization (WMO) is an intergovernmental organization that is dedicated to international cooperation and coordination on Earth’s atmosphere, its interaction with the land and oceans, the weather and climate it produces, and the resulting distribution of water resources.

² The National Aeronautics and Space Administration is America’s civil space program and the global leader in space exploration.

³ The United Nations is an international organization currently made up of 193 Member States, its work is guided by the purposes and principles contained in its founding Charter. The UN and associates have been awarded the Nobel Peace Prize more than ten times.

⁴ Eric Bender was an associate director for communications and public affairs at Whitehead Institute for Biomedical Research, founding editor in chief of MIT’s technologyreview.com and executive editor at pcworld.com. The container for this text is Knowable Magazine in the Annual Review section, which fact-checks and peer reviews its contents.

warming (Watson).⁵ Both Bender and Watson's suggestions on utilizing nature can be used to minimize the harmful effects of climate change on ocean ecosystems.

Oceans play a significant role in human society – from providing 15% of the animal protein intake for the world's population to being part of the livelihood for over 3 billion people – however, as climate change worsens, they are being severely affected (Ruckelshaus et al⁶; Ikeazor & Quickenborne⁷). Currently, ocean ecosystems are being threatened by issues such as a decrease in biodiversity and increases in both acidification⁸ and deoxygenation⁹ because of climate change (Levin & Bris).¹⁰ Without human intervention in aiding oceans to become more resilient to climate change, society may suffer greatly. Thus, it should be the utmost priority of international societies to do all they can to help sustain oceans.

Passing Policies to Protect Oceans

One possible solution is to increase the protections for oceanic territories. Oceanology experts explain that “[plans] to restrict direct human disturbance [...] may help to establish refugia for endangered species and habitats” and that “[protections should be implemented] to reduce physical and chemical disturbances [in order to] lessen chances of habitat loss and

⁵ Julia Watson is an architect, design activist and scholar who has taught at Harvard and Columbia University. She is an expert on Indigenous technologies, making her qualified to discuss them. This source is a TED talk, the company has fact-checkers and topic-specific advisors.

⁶ The fifteen scholars who worked on this paper have credentials in ocean-related studies or environmental science.

⁷ Sharon Ikeazor is the Nigerian Minister of State for the Environment. Vincent Van Quickenborne is the Deputy Prime Minister and Minister of Justice and the North Sea in Belgium.

⁸ As described by NOAA, ocean acidification refers to a reduction in the pH of the ocean over an extended period of time, caused primarily by uptake of carbon dioxide from the atmosphere. It harms calcifying organisms such as oysters, clams, sea urchins, corals, and calcareous plankton.

⁹ As described by the University of California San Diego, deoxygenation is the overall decline in the oxygen content of oceanic and coastal waters. It reduces the quality and quantity of suitable marine habitat, reduces growth rate, interferes with reproduction, and increases disease susceptibility.

¹⁰ Lisa A. Levin has worked for Center for Marine Biodiversity and Conservation at the University of California San Diego. Nadine Le Bris is an observatory oceanologist.

extinction of species” (Levin & Bris). Other environmental scientists agree with this solution, stating that “existing coastal habitat protection laws aimed at reducing eutrophication [can mitigate] processes that exacerbate ocean acidification” (Ruckelshaus et al). That being said, how should international societies decide which policies to implement, especially when over 80% of the world’s ocean is unobserved (NOAA^a)?¹¹ It is suggested that recognition of specific marine ecosystems by the United Nations Framework Convention on Climate Change can improve the research surrounding the ways in which ocean ecosystems help mitigate climate change (Levin & Bris). Bender’s paper demonstrates the benefits of analyzing specific parts of nature by discussing how the launch of the Global Urban Evolution Project has helped hundreds of scientists further understand the role of the white clover in urban ecosystems, which gives humanity a better understanding on how nature could be used to improve the world (Bender). However, this is easier said than done as “64% of the ocean is beyond national jurisdiction and [no one] is explicitly responsible for protecting [these marine] ecosystems”, so appointing specific societies to investigate and protect these areas is a limitation that prevents this from being the most effective solution (Levin & Bris).

Since protecting specific territories may be too difficult, what about fixing other factors that disrupt marine ecosystems such as overfishing? Scholars have connected climate change and overfishing to be the biggest difficulties that marine ecosystems face, with one paper reporting that the rapid rate of climate change on ocean ecosystems is so intense it increases the pressure on fish stocks to the point of potential extinction (Sumaila & Tai).¹² For instance, one study

¹¹ The National Oceanic and Atmospheric Administration holds key leadership roles in shaping international ocean, fisheries, climate, space and weather policies.

¹² U. Rashid Sumaila is professor and director for the Fisheries Economics Research Unit & the Ocean Canada Partnership at the University of British Columbia. Travis C. Tai is a marine fisheries scientist at UBC in the Institute for the Oceans and Fisheries.

found that “[in 2019,] at least 40 percent of fish stocks in the North East Atlantic and 87 percent in the Mediterranean and Black Seas [were] subject to unsustainable fishing practices” (Sumaila & Tai). This is a prevalent issue because it has been concluded by experts that “interactions between multiple human-induced stressors will continue to exacerbate [ecosystems and] progressively limit the adaptive capacity of natural systems to cope with rapid climate change” (Ling et al).¹³ Additionally, current global fishery management has been found to be incredibly inefficient, thus reforming fishing policies may be an effective solution (Sumaila & Tai).

Possible reforms include strategies that “implement management interventions for [endangered] species”, such as placing limitations on where fishing can occur to regulations on using certain fishing equipment (Pollom et al¹⁴; Monterey Bay Aquarium Seafood Watch^{a15}). Additionally, laws can be implemented to reduce illegal fishing as it has been cited to be one of the biggest contributing factors to overfishing (Monterey Bay Aquarium Seafood Watch^b). However, there is a limitation we must consider. Professionals in law and ocean studies have analyzed bills regarding oceans, such as the House Resolution 200¹⁶, and concluded that it would harm American fisheries, stating that passing this bill would “bring back unscientific management”, “risk the return of overfishing [in the US]”, and “make bycatch harder to measure and manage” (Carter & Polefka).¹⁷ House Resolution 200 was passed shortly after the aforementioned analysis was published and the effects of the legislation are still unclear (Congressional Record).¹⁸

¹³ All four scholars who worked on this paper have credentials in ocean-related studies and fields.

¹⁴ All twenty authors that contributed to this paper have done some kind of higher education and many of them have credentials in ocean-related studies.

¹⁵ Seafood Watch ranges from advising consumers about sustainable seafood to working directly with businesses and governments around the world about aquaculture and fisheries.

¹⁶ House Resolution 200 is a bill that proponents argue will benefit recreational saltwater anglers and commercial fishermen, with minimal costs. For instance, this bill requires the Gulf States Marine Fisheries Commission to conduct all fishery stock assessments used for management purposes by the Gulf of Mexico Fishery Management Council.

¹⁷ Alexandra Carter is the former deputy director of Ocean Policy at the Center for American Progress. Shiva Polefka is an associate director for Ocean Policy at the Center for American Progress.

¹⁸ Congress.gov is the official website for U.S. federal legislative information. The site provides access to legislative information for Members of Congress, legislative agencies, and the public.

However, this demonstrates the possibility of ineffective laws being passed, and this risk factor prevents this solution from being ideal.

Using the Ocean Itself to Combat Global Warming

In Watson's presentation, she describes how Calcutta's water supply is often polluted by trash from other cities; 300 fish ponds are used instead of floodplains to clean the water (Watson). Watson advocates for an increased focus on using natural technologies rather than man-made ones, thus in the context of this paper, focusing on oceans themselves may create successful methods that combat climate change. Many ocean ecosystems have naturally occurring effects that can serve as solutions towards climate change if accentuated. The method of iron fertilization, as explained by the Woods Hole Oceanographic Institution, stimulates the natural process of phytoplankton¹⁹ undergoing photosynthesis to convert carbon dioxide into oxygen (WHOI).²⁰ Phytoplankton need only small amounts of iron to grow into large blooms, the likes of which can "be seen from space" (WHOI). It has been shown to remove large amounts of carbon dioxide from the atmosphere, an example being how "[soot] from the 2019-2020 Australian wildfires [caused a phytoplankton bloom that may have removed 150 to 300 million tons of carbon from the atmosphere]" (WHOI). This is done through the aforementioned photosynthesis, along with "[the phytoplankton taking] up extra carbon [which then] would sink into the depths [of the ocean]" (Cornwall).²¹ However, there are many limitations that prevent the implementation of iron fertilization as a main solution. The success rates of this process are low and researchers are uncertain of how big of an impact it actually makes nor what the possible

¹⁹ As described by Oxford Languages, phytoplankton are plankton consisting of microscopic plants.

²⁰ Woods Hole Oceanographic Institution is the world's leading, independent non-profit organization dedicated to ocean research, exploration, and education.

²¹ Warren Cornwall is an environmental, science and outdoor recreation journalist whose stories have appeared in many new outlets such as Science.

side effects are (Cornwall). According to marine radiochemist Ken Buesseler, out of “13 large iron fertilization experiments that boosted algae growth, only two demonstrated removal of carbon to the deep sea” (Underwood).²² Additionally, Dave Siegel, a marine scientist at the University of California, states that the amount of “absorbed carbon [that] makes it to the deep ocean is uncertain” as other factors can re-emit the carbon, and that it may be difficult “to track these carbon flows [accurately]” (Cornwall). Furthermore, there are concerns about possible negative impacts “such as toxic algal blooms”, which further contribute to deoxygenation and the disruption of ecosystems (Underwood; Cornwall). Successfully implementing iron fertilization is extremely unlikely due to its many flaws, preventing it from being an ideal solution (Denman).²³

Going back to Watson’s TEDtalk, she describes how indigenous populations have used Qasab reeds for food, animal feed, and construction material (Watson). Again, this demonstrates how nature can be utilized to improve life; in the case of oceans, kelp forests can be used to sequester carbon and improve agricultural practices. Kelp forests are dense groups of large brown algae that grow relatively close to the surface, providing food and shelter for many aquatic organisms (NOAA^b). It isolates carbon by exporting macroalgae²⁴ “to the deep sea [where] the carbon from [the] macroalgae is stored far away from the shore, [thus being] less likely to be disturbed and returned to the atmosphere” (Hurlimann & Zucker).²⁵ These

²² Emily Underwood is a writer for Science. She has a master’s degree in science writing from Johns Hopkins University, as well as a bachelor’s degree in Science and Technology Studies with a focus in neuroscience from Brown University.

²³ Kenneth L. Denman is a professor in the School of Earth and Ocean Sciences at the University of Victoria, and Senior Scientist with the VENUS Coastal Network. He received a PhD in ocean physics from the University of British Columbia.

²⁴ Macroalgae refers to thousands of species of macroscopic, multicellular, marine algae. They are primary producers that form the basis of many marine food chains.

²⁵ During the time the article was written, Sylvia Hurlimann was a third-year graduate student in the Department of Molecular and Cellular Biology at Harvard University. Hannah Zucker was a second-year PhD candidate in the Program in Neuroscience at Harvard University.

macroalgae are thought to be closed off from the atmosphere for potentially millions of years, thus giving humanity more time to tackle global warming (Hurlimann & Zucker). In a study conducted by professionals in ecology, it was found that “farming seaweed²⁶ in just 3.8% of federal waters off the Californian coast could offset all the carbon emissions from the state's [50 billion dollar] agriculture industry [and] that 48 million sq km of the world's oceans are suitable for seaweed cultivation” (Gerretsen).²⁷ In addition, seaweed can be used in both aquatic and terrestrial farms to reduce the amount of harmful feeds in fisheries and methane – a main source of greenhouse gas emissions – in cattle feeds (Filbee-Dexter).²⁸ Yet, when putting this solution into practice, a couple of limitations appear. The current research on kelp forests is lacking as “[many] regions are missing baseline data for detecting change”, with “[large errors surrounding the] estimates of associated [impacts]” (Filbee-Dexter). Furthermore, as discussed by marine ecologist Jordan Hollarsmith, it may be difficult for researchers to track kelp that drifts off, which could release carbon back into the atmosphere if decomposed (Bryce).²⁹ There is also a lack of research into the “ecological impact of depositing billions of tons of dead biomass on sea floor”, making this solution currently unviable (Temple).³⁰

Increase Focus on Ocean Based Renewable Energies

The aforementioned solutions all had flaws that prevented them from being an ideal solution, so then, what is? Through analyzing the arguments made by Eric Bender and Julia

²⁶ For clarification, seaweed is a term used to describe many different marine-based species of algae, and kelp describes the largest subgroup of seaweed.

²⁷ Isabelle Gerretsen is a graduate of City Journalism School and the University of Oxford who works as a reporter for Climate Home News and a writer and editor for BBC Future.

²⁸ Karen Filbee-Dexter is a marine ecologist at the University of Western Australia.

²⁹ Emma Bryce is a freelance journalist based in London, one of the topics she writes about is the environment. Her work has appeared in news outlets such as the Guardian.

³⁰ James Temple is the senior editor for energy at MIT Technology Review, focusing on renewable energy and the use of technology to combat climate change.

Watson, the ideal solution must utilize existing natural resources that have been continuously proven to work effectively without glaring limitations. Thus, this paper argues that the most effective method for international societies in aiding oceans in the face of the climate crisis is increasing the usage of ocean based renewable energy. These include wind, wave, ocean thermal and tidal energy among others (Huckerby et al).³¹ Ocean-based technologies generate renewable energies that have almost no carbon emissions, “thus [are] able to displace emissions associated with fossil-based electricity generation” (Hoegh-Guldberg et al).³² For instance, “wind turbines [can generate off-shore wind power which] mechanically turn electric generators” (United Nations^b). Wind energy has been improved upon throughout the years and may potentially “cover more than one third of global power needs” (United Nations^b). Scientists have also used sea water to generate “kinetic and thermal energy [which produces] electricity or heat” without contributing to the climate crisis (United Nations^b). Although there are no doubt many benefits in using ocean based energies, there are potential limitations in furthering its development that must first be addressed. A paper written by engineering professionals discussed the environmental impacts of renewable energy source based electrical power plants, including ocean related ones (Rahman et al).³³ For example, although tidal energy can be easily predicted unlike other sources of renewable energy, “very few locations demonstrate promising potential” and the turbines used could “create turbulence, vortices, sedimentation, and turbidity” (Rahman et al). Additionally, the construction of tidal barrages may be too costly, as La Rance, one of the first ever built tidal barrages, had an initial investment of approximately 100 million US dollars (Rahman et al).

Wave and ocean thermal energy are also potentially dangerous as the devices submerged in

³¹ This paper was written under the Intergovernmental Panel on Climate Change, which is an intergovernmental body of the United Nations whose job is to advance scientific knowledge about climate change caused by human activities.

³² This paper was written under the Ocean Panel, also known as the The High Level Panel for a Sustainable Ocean Economy, which is a global initiative by serving world leaders that works towards a sustainable ocean economy.

³³ All three authors who worked on this paper have credentials in engineering and technology.

oceans to gather these energies may cause pollution, and possible chemical spillage can severely toxify surrounding areas (Rahman et al). However, it must be recognized that these concerns are only hypothetical and further research into pursuing ocean based energies can lower risk rates. There already exist government policies that support this solution, such as giving “capital grants to device developers; performance incentives for produced electricity; marine infrastructure development; standards, protocols and regulatory interventions for permitting; and space and resource allocation” (Huckerby et al). If societies around the world all contribute to advancing the research and development of ocean based energies – which includes “[improving] performance, reliability, and durability, while reducing costs” –, it is possible to make the current theoretical potential of ocean energies a reality: that the amount of energy produced will “well [exceed] current and future human energy needs” with minimal damage on the environment (Hoegh-Guldberg et al;Huckerby et al).

In the face of the ongoing climate crisis, it is of the utmost importance to protect oceans and the ecosystems within them, as all human societies rely on them to some extent in order to survive and flourish. It is the responsibility of all nations to aid oceans in becoming resilient to climate change and advance the usage of renewable ocean resources in order to prevent humanity from facing further destruction. After all, oceans have provided so much for the world, it is only correct to help it become as resilient as it helped humanity become.

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